

PHASE 1: PROJECT PLANNING & ANALYSIS

(Week 1-2)

Step 1: Initial Project Setup

- ☐ Create project repository (GitHub/GitLab)
- ☐ Set up team communication channels (Discord/Slack)
- ☐ Schedule regular team meetings
- ☐ Assign primary and backup roles for each task

Step 2: System Analysis Documentation (SAD)

2.1 Brief Description of the System - Kim

2.2 Problem Analysis - Wes

2.3 SWOT Analysis - Angelica

Create a 2x2 matrix:

- ☐ **Strengths:** What advantages does your LMS have?
- ☐ **Weaknesses:** What limitations exist?
- ☐ **Opportunities:** Market trends, user needs
- ☐ **Threats:** Competition, technical challenges

2.4 Requirements Gathering

Functional Requirements - Julia List what the system **MUST** do:

Non-Functional Requirements - Kim

MoSCoW Analysis - Andrew Prioritize features:

PHASE 2: SYSTEM DESIGN (Week 3-4)

Step 3: Frontend Design Diagrams

3.1 Use Case Diagram - Leanne

3.2 Class Diagram - Leanne, Wes

3.3 Activity Diagram - Kim

3.4 System Flowchart - Kim, Andrew

Step 4: Backend Design Diagrams

4.1 Entity Relationship Diagram (ERD) - Angelica

Define database entities:

- [] **Users** (userID PK, name, email, password, role, createdAt)
- [] **Courses** (courseID PK, title, description, instructorID FK, createdAt)
- [] **Enrollments** (enrollmentID PK, userID FK, courseID FK, enrollmentDate)
- [] **Modules** (moduleID PK, courseID FK, title, order)
- [] **Content** (contentID PK, moduleID FK, type, fileURL, title)
- [] **Assignments** (assignmentID PK, courseID FK, title, dueDate, points)
- [] **Submissions** (submissionID PK, assignmentID FK, userID FK, grade, submittedAt)
- [] **Payments** (paymentID PK, userID FK, courseID FK, amount, status, date)
- [] Show relationships (one-to-many, many-to-many)

4.2 Data Flow Diagram (DFD) - Kim

Create Level 0 and Level 1 DFDs:

4.3 Database Schema - Angelica

- [] Write CREATE TABLE statements for all entities
- [] Define primary keys, foreign keys
- [] Add indexes for performance
- [] Define constraints (NOT NULL, UNIQUE)

PHASE 3: UI/UX PROTOTYPE (Week 4-5)

Step 5: Design Prototype - TEAM EFFORT

Design Principles:

PHASE 4: DEVELOPMENT (Week 6-10)

Step 6: Backend Development - Angelica, Andrew

6.1 Setup Development Environment

- [] Install XAMPP/WAMP (Apache, MySQL, PHP)
- [] Set up database in phpMyAdmin
- [] Create project folder structure

6.2 Database Implementation

6.3 Core Backend Features

- [] **Authentication System**
 - Registration endpoint (POST /api/register.php)
 - Login endpoint (POST /api/login.php)
 - Session management
 - Password hashing (password_hash, password_verify)
- [] **User Management API**
 - Get user profile (GET /api/users/{id})
 - Update profile (PUT /api/users/{id})
 - Delete user (DELETE /api/users/{id})
- [] **Course Management API**
 - Create course (POST /api/courses)
 - Get all courses (GET /api/courses)
 - Get course by ID (GET /api/courses/{id})
 - Update course (PUT /api/courses/{id})
 - Delete course (DELETE /api/courses/{id})
- [] **Enrollment API**
 - Enroll student (POST /api/enrollments)
 - Get user enrollments (GET /api/enrollments/user/{id})
 - Unenroll (DELETE /api/enrollments/{id})

- ☐ **Assignment & Submission API**
 - Create assignment (POST /api/assignments)
 - Submit assignment (POST /api/submissions)
 - Grade submission (PUT /api/submissions/{id}/grade)
- ☐ **File Upload Handler**
 - Handle course materials upload
 - Handle assignment submissions
 - Validate file types and sizes

Step 7: Frontend Development - Leanne, Julia, Wes

7.1 Setup Frontend

7.2 Build Pages

- ☐ **Landing Page**
- ☐ **Authentication Pages**
- ☐ **Dashboards**
- ☐ **Course Pages**
- ☐ **Interactive Features**

7.3 JavaScript Functionality

PHASE 5: SYSTEM INTEGRATIONS (SIA) (Week 8-11)

Step 8: Email Integration - Angelica

Using PHPMailer:

- ☐ Install PHPMailer library
- ☐ Configure SMTP settings (Gmail/SendGrid)
- ☐ Create email templates
- ☐ Implement email functions:
 - Welcome email on registration
 - Course enrollment confirmation

- Assignment due date reminders
- Grade notification
- Password reset email

Step 9: SMS Integration - Andrew

Step 10: AI Integration - Angelica

Possible AI Features:

- **[] Chatbot for Student Support**
 - Use OpenAI API or similar
 - Answer FAQs
 - Guide navigation
- **[] Content Recommendations**
 - Suggest courses based on interests
 - Recommend related materials
- **[] Automated Grading (optional)**

Step 11: Payment Integration - Julia

Using PayPal/Stripe:

Step 12: RESTful API with Partner Group - Wes

STUDENT

Can Do:

- Register/Login
 - Browse course catalog
 - Search/filter courses
 - View course details
 - Enroll in courses
 - Pay for paid courses
 - Access enrolled courses
 - Watch videos
 - Download course materials (PDFs, documents)
 - Take quizzes
 - Submit assignments
 - View grades for specific course
 - Track course progress
 - Receive notifications (email/SMS)
 - Message teacher
 - Rate and review courses
 - View transaction history
 - Update profile
 - Change password
 - E- Certificate When course is done.
 - Ask the chatbot (AI)
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TEACHER

Can Do:

- Register/Login as instructor
- Create new courses
- Edit/delete courses
- Upload course content (videos, PDFs, presentations)
- Organize content into modules/sections
- Create quizzes (multiple choice, true/false)
- Create assignments
- Set due dates
- Grade assignments manually
- Provide feedback on submissions
- View student submissions
- Download submitted files
- View enrolled students list
- Track student progress
- Send announcements to students
- Message students
- View course analytics (enrollments, completion rates)
- Set course price
- View earnings
- Update profile

ADMIN

Can Do:

- Login to admin panel
- View dashboard (total users, courses, revenue)
- Manage users:
 - View all students
 - View all teachers
 - Activate/deactivate accounts
 - Delete users
 - Reset passwords
 - Can Update Users Info.
- Manage courses:
 - View all courses
 - Approve/reject courses
 - Feature courses
 - Delete courses
 - Edit course details
- View all transactions
- Generate reports:
 - User statistics
 - Course statistics
 - Revenue reports
 - Enrollment reports
- Manage categories
- Manage payments
- Process refunds
- View and resolve disputes
- Configure system settings
- Send system-wide announcements
- View activity logs
- Moderate reviews/comments
- Manage platform commission rates